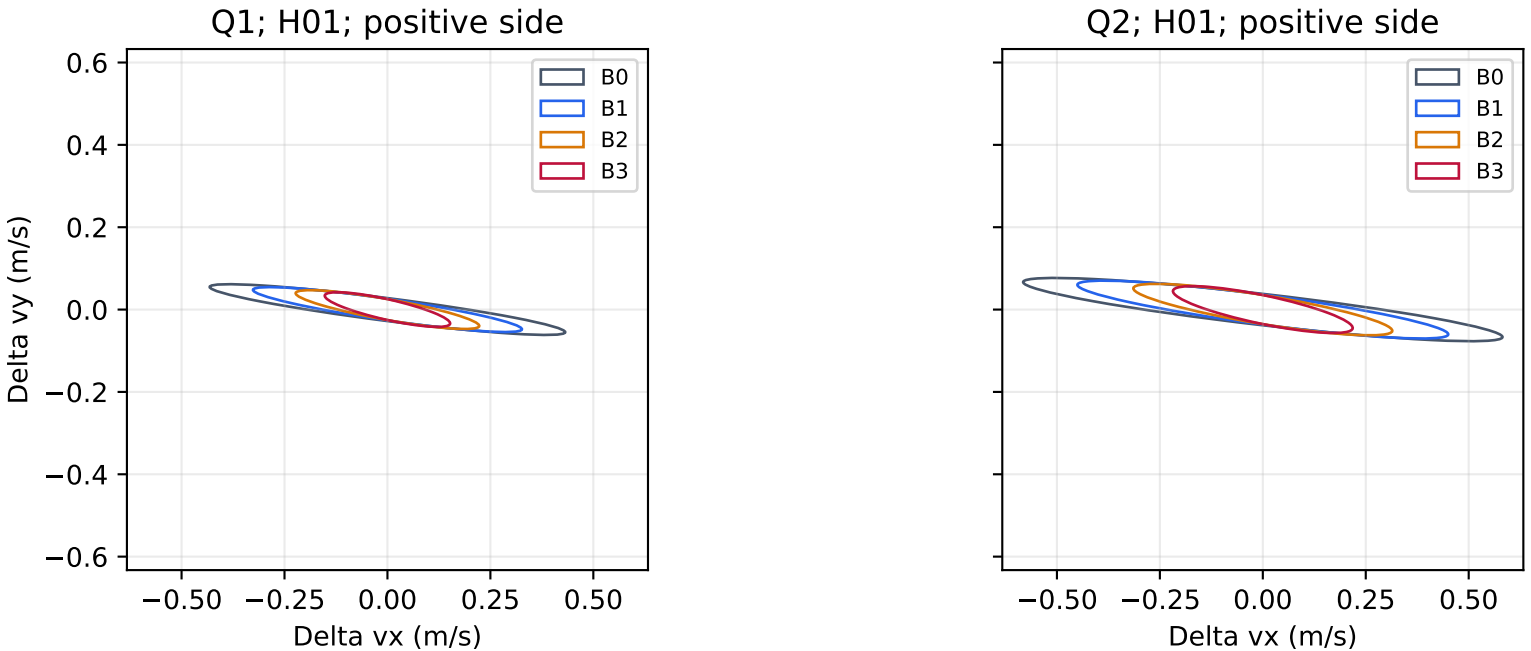Profiled local1-sigma velocity ellipses; unknown position and bias